

Design Documentation — Marketing Impulso

Reporting Dashboard

Prepared for: Google Ads API — Basic Access Application **Applicant:** Marketing Impulso **MCC ID:** 825-993-9796
Date: May 2026 **Permissible Use Requested:** Reporting (read-only)

1. Executive Summary

Marketing Impulso is a Brazilian digital marketing agency. We are building an **internal reporting dashboard** ("Plataforma Impulso") used exclusively by our own staff to monitor paid-media performance across the client accounts linked to our manager account (MCC 825-993-9796).

The tool is **not** distributed to clients, **not** white-labeled, **not** resold, and performs **no write operations** on Google Ads accounts. All API calls are read-only reporting queries (`GoogleAdsService.SearchStream` and `account_budget` resource via GAQL).

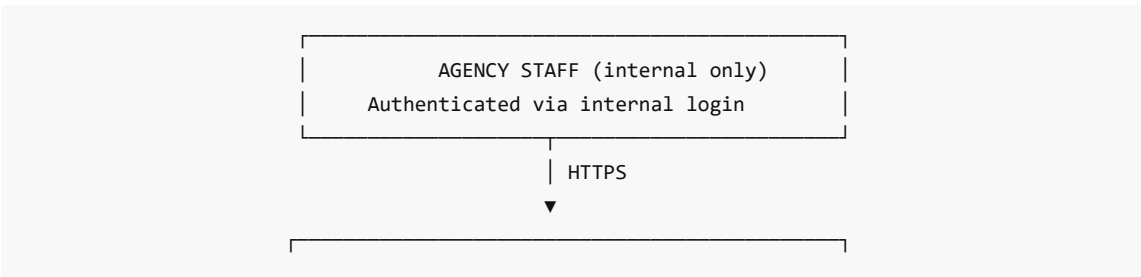
This document describes the architecture, data flows, security posture, rate-limit strategy, and compliance controls of the application.

2. Business Context

Item	Value
Company	Marketing Impulso
Country	Brazil
Business model	Digital marketing agency (paid media management)
Number of staff	Small team (< 10 internal users of the tool)
Number of client Google Ads accounts	~10–30 (all linked to MCC 825-993-9796)
Client authorization	Written authorization on file for every linked account
Public website	https://marketingimpulso.com
API contact email	vxdigitalbusiness@gmail.com

3. System Architecture

3.1 High-level diagram





3.2 Components

Component	Technology	Purpose
Web frontend	Next.js 15 (App Router) + TypeScript	Renders the dashboard UI for internal staff
ORM	Prisma	Type-safe access to PostgreSQL
Database	PostgreSQL (managed on EasyPanel)	Stores client registry and cached metrics
Orchestration	n8n (self-hosted)	Schedules and executes API sync workflows
Auth (staff)	Internal email/password with session cookies	Restricts dashboard to agency employees
Hosting	EasyPanel VPS (Brazil region)	Single-tenant deployment

Important: The Next.js application **never calls the Google Ads API directly**. All API access is funneled through n8n workflows, which makes rate-limit accounting and credential isolation straightforward.

4. Google Ads API Usage

4.1 Endpoints and resources used

Endpoint / Resource	Operation	Frequency	Purpose
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customers/{customer_id}/googleAds:searchStream	Read	Hourly, 08:00– 19:00, Mon–Fri (BR time)	Fetch account budget and currency for each linked client
account_budget resource (via GAQL)	Read	(same as above)	Pull approved_spending_limit_micros, amount_served_micros, status
customer.currency_code	Read	(same as above)	Resolve currency for correct balance formatting

4.2 Example GAQL query

```

SELECT
  customer.currency_code,
  account_budget.approved_spending_limit_micros,
  account_budget.amount_served_micros
FROM account_budget
WHERE account_budget.status = 'APPROVED'

```

4.3 Operations per day (estimate)

- **Linked clients:** 30 (upper bound)
- **Sync windows per day:** 12 (hourly, 08:00–19:00 BR)
- **Queries per sync per client:** 1
- **Estimated daily operations:** $30 \times 12 = 360 \text{ ops/day}$

This is well under the Basic Access daily quota of 15,000 operations.

4.4 Operations NOT performed

The following are **explicitly out of scope** for this application:

- No `MutateOperation` calls of any kind
- No campaign, ad group, ad, asset, or keyword creation
- No bid or budget modification
- No account creation, linking, or unlinking
- No conversion uploads
- No customer match / audience uploads
- No Smart Bidding / experiment operations

5. Authentication & Authorization

5.1 OAuth 2.0 flow (Google Ads API)

1. The MCC owner (`wxdigitalbusiness@gmail.com`) completes the OAuth 2.0 consent flow once, granting scope `https://www.googleapis.com/auth/adwords` .
2. The resulting **refresh token** is stored encrypted as a server-side secret (n8n credential vault / environment variable). It is **never** exposed to the browser or to client accounts.

3. Each scheduled sync exchanges the refresh token for a short-lived access token via `https://oauth2.googleapis.com/token`.
4. The access token is used only for the duration of that sync run (typically < 60 seconds) and discarded.

5.2 Internal staff authentication

- Agency staff authenticate to the dashboard via email/password.
- Sessions use HTTP-only, secure, SameSite=Lax cookies.
- The dashboard is reachable only by authenticated staff; there is no public/anonymous view.
- The application is **single-tenant** (Marketing Impulso only). External clients do not have accounts.

5.3 Authorization between staff and client data

- Every linked Google Ads account in our MCC has documented written authorization from the advertiser.
 - All staff currently have full read access to all client metrics inside the dashboard, since they all work on these accounts in the normal course of business. Role-based access control may be added later.
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6. Data Storage & Retention

6.1 What we cache locally

The PostgreSQL `clientes_ativos` table stores, **per client**:

- `google_ad_customer_id` — the linked Google Ads customer ID
- `ultimo_saldo_google` — last fetched account balance (numeric)
- `ultimo_tipo_conta_google` — account billing type label
- `saldo_google_atualizado_em` — timestamp of last sync
- `ultimo_erro_google` — last API error message, if any
- `limite_minimo_google` — internal alert threshold (not from Google)

We do **not** cache personally identifiable information (PII) about end-users of the advertiser, conversion data, audience lists, or any data subject to the Customer Match policy.

6.2 Retention

- Cached metrics are overwritten on every sync (1-hour cadence) — no historical aggregation is stored long-term in v1.
- Alert logs (`alertas_saldo_log`) are retained for operational audit purposes for 90 days, then pruned.

6.3 Data residency

All cached data resides in PostgreSQL hosted on EasyPanel infrastructure in Brazil.

7. Rate Limiting & Error Handling

7.1 Self-imposed limits

- Hard cap: 1 query per client per sync window (no fan-out).
- Sync windows are spaced 1 hour apart and limited to business hours (Mon–Fri, 08:00–19:00 BRT).
- Maximum daily operations remain at ~360 ($\approx$ 2.4% of the Basic Access daily quota).

7.2 Error handling

- Transient errors (5xx, network timeouts): logged to `ultimo_erro_google`, no retry within the same hour. Next scheduled run retries naturally.

- `QuotaError` / `RESOURCE_EXHAUSTED` : workflow halts and emits an internal alert; we do not auto-retry to avoid amplifying quota pressure.
- `AuthenticationError` / `AuthorizationError` : workflow halts; staff is notified to re-consent.
- Errors are persisted per-client so the dashboard can surface "stale data" indicators to staff.

7.3 Backoff strategy

Because each client is queried at most once per hour, no in-loop retries or aggressive backoff are required. The hourly cron provides natural spacing.

8. Security

Control	Implementation
Developer token storage	Server-side environment variable (n8n credentials vault). Never sent to browser.
OAuth refresh token storage	Server-side encrypted credential in n8n. Never logged.
Transport	All API traffic over HTTPS/TLS 1.2+
Database access	Only the n8n service and the Next.js server can connect; no public ingress.
Staff login	HTTP-only secure cookies; passwords hashed with bcrypt (cost ≥ 10)
Audit logging	All scheduled API calls log a row with client ID, timestamp, status, and error (if any)
Source control	Private repository; secrets excluded via <code>.gitignore</code>
Secret rotation	OAuth refresh token and developer token are rotatable from the operator console without redeploy

9. Compliance with Google Ads API Policies

We have reviewed and will comply with:

- **API Terms of Service** — accepted by MCC owner
- **Required Minimum Functionality (RMF)** — N/A for read-only Reporting use case, but acknowledged
- **System Name disclosure** — every request sets a stable `User-Agent` identifying "Plataforma Impulso / Marketing Impulso"
- **Permissible Use** — application is exclusively Reporting; no Ad Management or Keyword Research operations
- **Data handling** — we do not redistribute Google Ads data to third parties; data stays inside the agency
- **Customer Match / sensitive data** — not used

10. Roadmap (out of scope for current Basic Access request)

The following are *possible* future capabilities. If/when we add any of them, we will re-apply for the appropriate access level (e.g., Standard Access for unlimited operations or for any write-side functionality):

- Campaign performance time-series (would still be read-only)
- Multi-currency consolidated reporting

- Daily PDF reports emailed to internal account managers

We are **not** requesting access for these capabilities at this time.

11. Contact

Company	Marketing Impulso
Website	https://marketingimpulso.com
API contact email	wxdigitalbusiness@gmail.com
MCC ID	825-993-9796
Country	Brazil

End of document.